STAT 405 - Project Report

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Data

The Vancouver Police Department has been counting crime counts for specific neighborhoods, the publically availble crime data is presented in an pdf found at:

`https://vpd.ca/wp-content/uploads/2026/01/neighbourhood-stats-year-end-2025.pdf`

The dataset contains reported counts of several crime categories across Vancouver neighbourhoods, including sex offences, assaults, robbery, break and enter, theft of motor vehicle, theft from auto, theft under $5000, arson, mischief, and offensive weapons.

Because the data are published in PDF format rather than as a directly downloadable CSV, we manually entered the table into an R dataframe.

In our analysis we hope to use Bayesian analysis to quantify crime rates for different types of crimes, and whether we can predict future crime counts for selected neighbourhoods and crime types.

The motivation behind our study is to hopefully find ways in which crime can be prevented, for instance if we find that crime rates are higher in a specific neighbourhood then crime rates could be decreased by enforcing stricter policing in that area.

```r
crime_data <- data.frame(
  neighbourhood = c(
    "Arbutus Ridge","Central Business District","Dunbar-Southlands","Fairview",
    "Grandview-Woodland","Hastings-Sunrise","Kensington-Cedar Cottage",
    "Kerrisdale","Killarney","Kitsilano","Marpole","Mount Pleasant","Musqueam",
    "Oakridge","Renfrew-Collingwood","Riley Park","Shaughnessy","South Cambie",
    "Stanley Park","Strathcona","Sunset","Victoria-Fraserview","West End",
    "West Point Grey"),
  sex_offences = c( 0,151,8,16,27,29,34,6,14,12,8,24,2,8,30,12,0,13,1,43,24,15,43,4),
  assaults = c(18,1591,22,115,254,97,185,22,51,101,69,215,15,26,168,41,15,24,18,622,80,
               45,306,24),
  robbery = c( 1,218,0,9,22,7,26,3,3,10,4,36,0,0,22,3,2,2,4,64,9,9,35,2),
  break_enter = c(25,438,24,164,121,84,127,29,37,92,89,173,0,44,123,75,32,33,12,
                  151,77,29,149,29),
  theft_mv = c(7,74,7,16,64,31,48,10,24,18,22,28,0,7,53,13,6,6,2,48,36,24,23,5),
  theft_auto = c(41,1855,72,219,219,222,317,73,84,202,146,220,0,56,202,148,43,
                 56,54,369,148,136,443,60),
  theft_under_5k = c(69,4458,70,706,507,312,601,132,159,465,276,1015,
                     1,115,1446,178,38,140,53,603,637,125,1120,77),
  arson = c(1,68,1,6,15,9,8,0,1,9,1,20,0,0,8,5,0,2,2,37,4,3,10,0),
  mischief = c(34,1864,33,209,281,140,209,45,83,142,134,296,6,29,171,86,28,42,16,434,137,
               85,318,32),
  offensive_weapons = c(4,708,4,28,44,9,25,5,5,10,8,39,0,1,9,2,3,4,4,244,9,10,38,3))
```

```r
head(crime_data)
```

```
##                neighbourhood sex_offences assaults robbery break_enter theft_mv
```

```
## 1             Arbutus Ridge            0       18       1          25        7
## 2 Central Business District          151     1591     218         438       74
## 3         Dunbar-Southlands            8       22       0          24        7
## 4                  Fairview           16      115       9         164       16
## 5        Grandview-Woodland           27      254      22         121       64
## 6          Hastings-Sunrise           29       97       7          84       31
##   theft_auto theft_under_5k arson mischief offensive_weapons
## 1         41             69     1       34                 4
## 2       1855           4458    68     1864               708
## 3         72             70     1       33                 4
## 4        219            706     6      209                28
## 5        219            507    15      281                44
## 6        222            312     9      140                 9
```

Models

Since the data are counts, we model y_{ij} using Poisson distributions to estimate underlying crime intensity.

We consider two models.

Model 1: Simpler model

We treat each neighbourhood-crime combination independently:

$$y_{ij} \mid \lambda_{ij} \sim \text{Poisson}(\lambda_{ij}), \quad \lambda_{ij} \sim \text{Gamma}(\alpha, \beta)$$

The Gamma prior is conjugate to the Poisson likelihood, allowing straightforward posterior computation.

Model 2: Hierarchical model

We incorporate neighbourhood and crime-type effects:

$$y_{ij} \mid \lambda_{ij} \sim \text{Poisson}(\lambda_{ij}), \quad \log(\lambda_{ij}) = \mu + \alpha_i + \beta_j$$

$$\alpha_i \sim \mathcal{N}(0, \sigma_\alpha^2), \quad \beta_j \sim \mathcal{N}(0, \sigma_\beta^2)$$

where μ is an overall intercept, α_i is the neighbourhood effect, and β_j is the crime-type effect.

We use Gaussian priors for α_i and β_j, reflecting that most effects are near zero unless supported by the data. The variance parameters σ_α^2 and σ_β^2 control variability across groups.

To ensure identifiability, we impose

$$\sum_i \alpha_i = 0, \qquad \sum_j \beta_j = 0.$$

We interpret λ_{ij} as the expected number of crimes per year for crime type j in neighbourhood i, representing a yearly rate without explicit adjustment for exposure or opportunity.